Reproducing MG-GAT: Interpretable Rating Prediction with Geometric Deep Learning

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Paper: Leng, Liu & Ruiz — *Interpretable Recommendations and User-Centric Explanations with Geometric Deep Learning* (SSRN 3696092)
Task: Stage 1 only (rating prediction, no LLM). Target: Test RMSE ≈ 1.249 [Table 1, PA]

1. Problem & Scope

MG-GAT is a Multi-Graph Graph Attention Network for **matrix completion**: predicting how a user would rate a business they have not visited. The user-business rating matrix X is extremely sparse (fill rate < 0.012%). The model uses side features (S_u, S_b) and graph structures (G_u, G_b) to overcome this sparsity.

Per assignment requirements, all LLM components were excluded: the 4th business graph (LLM perceptual map), Algorithm 2 (explanation generation), and Section 4 of the paper. Only 3 business graph types were retained. [Section 3.2; Appendix D1]

2. Data

2.1 Dataset Version & Scale

The original paper used the 2019 Yelp dataset (no longer publicly available). This work uses the 2024 version, which is significantly larger but sparser.

Statistic	This Work	Paper (Table D1)	Ratio
Users	320,212	76,865	4.2×
Businesses	31,663	10,966	2.9×
Total reviews	1,183,126	260,350	4.5×
Avg reviews/user	3.70	3.387	—
Avg user friend degree	5.449	5.557	—

2.2 Time Split [Section 5.1; Appendix D2]

Split	Period	Reviews
Train	2009–2016	820,496
Validation	2017	179,662
Test	2018	182,968

Pipeline: filter PA businesses from `business.json` → filter reviews → extract users → extract features from `user.json` + `checkin.json` → split by year.

2.3 Features

User features Su (65 dims = 33 explicit + 32 implicit SVD) [Table D4]: compliments (11), votes (3), fans (1), yelping_since year/month/day (3), elite 2005–2018 + None (15). Min-max normalised.

Business features Sb (1,428 dims = 1,396 explicit + 32 implicit SVD) [Table D5–D6; Appendix D1]: attributes 87 (paper: 93), categories 1,149 (paper: 946) — differences due to dataset version — hours 14, check-in 144, location 2. Paper's explicit total: 93+946+14+144+2 = 1,199. [Appendix D1]

Implicit features [Appendix D1]: Binarise training ratings → truncated SVD (k_i=32) → $S_{u,imp} = U^{(0)}\Sigma^{1/2}$, $S_{b,imp} = B^{(0)T}\Sigma^{1/2}$ → concatenate with explicit features. SVD uses **training data only** to prevent leakage. Non-predictive implicit features are automatically suppressed via L2 regularisation.

2.4 Graphs

Graph	Construction	k	Reference

G_u (friendship)	Yelp friends field, undirected	—	Appendix D1
G_b geo	Haversine distance, k-NN	10	Appendix D1
G_b co-visit	Shared customers (train only)	10	Appendix D1
G_b category	Shared category count	10	Appendix D1

Three graphs are stored independently and combined via learnable ω_g weights in Eq.3 — they are **not** merged into a single adjacency matrix. [Section 3.3, Eq.3]

3. Model Architecture

3.1 Five-Layer Network [Section 3.3, Eq.2–6]

Layer	Equation	Purpose	Key Detail
1 — Linear	$H^{(1)} = W^{(1)} \cdot S$	Project raw features	No bias, no activation — preserves FR interpretability [Eq.2]
2 — Attention (NIG)	$\alpha = \text{softmax}(\text{LReLU}(a^T \cdot [H^{(1)}_i \parallel H^{(1)}_k]))$	Compute neighbor importance	Business side: 3 graphs weighted by ω_g, unified softmax over all neighbors [Eq.3]
3 — Aggregation	$H^{(2)}_i = \Sigma_{k \in Ni}\, \alpha \cdot H^{(1)}_k$	Aggregate neighbor info	Neighbours only, excludes self [Eq.4]
4 — Nonlinear	$H^{(3)} = \text{actv1}(W^{(2)} \cdot H^{(2)} + W^{(2)}_s \cdot S + b)$	Nonlinear transform + skip	Skip connection from raw features S [Eq.5]
5 — Final	$U_i = \text{actv2}(W^{(3)} \cdot H^{(3)}_i) + H^{(4)}_i$	Final embedding	$H^{(4)}$ = free embedding, graph-Laplacian regularised [Eq.6]

3.2 Prediction & Loss

Rating prediction [Eq.7]: $\hat{X}_{ij} = (r_{max} - r_{min}) \cdot \sigma(U_i \cdot B_j^T + b_{u,i} + b_{b,j} + b) + r_{min} = 4 \cdot \sigma(\ldots) + 1$

Loss [Eq.8]: $L = \|\Omega_{train} \circ (X - \hat{X})\|^2 + \theta_1 \cdot [\text{Tr}(H^{(4)T}_u \tilde{L}_u H^{(4)}_u) + \text{Tr}(H^{(4)T}_b \tilde{L}_b H^{(4)}_b)]$ where $\tilde{L} = L + \theta_2 \cdot I$

Key constraints: Graph regularisation applies **only to H(4)**, not $H^{(3)}$ — because Layers 2–3 already enforce local smoothness via attention [Section 3.4]. Loss is computed only at observed rating positions. All parameters have L2 regularisation (weight_decay). Optimiser: Adam + ReduceLROnPlateau. [Footnote 6]

3.3 Interpretability [Definitions 1–2]

Feature Relevance (FR): $FR = a^T \cdot W^{(1)}$ — directly readable because Layer 1 is linear. **Neighbor Importance Graph (NIG)**: attention weights α show which neighbors matter most; ω_g show relative graph-type importance.

4. Hyperparameter Search

The paper uses Hyperopt (TPE) but does not disclose final values [Footnote 6]. We searched 20 trials × 30 epochs:

Parameter	Search Space
θ_1	{0.001, 0.01, 0.1}
θ_2	{0.001, 0.01, 0.1}
kf	{32, 64, 128}
d0	{32, 64}
d1	{64, 128}
lr	{0.001, 0.005}
actv1	{elu, relu, tanh}

Best parameters found:

Parameter	Value

θ_1, θ_2	0.1, 0.1
kf, d0, d1	128, 64, 64
lr	0.005
actv1, actv2	relu, relu
weight_decay	1e-5

5. Experimental Results

5.1 Experiment Progression

Exp	Key Change	Val RMSE	Test RMSE	Train-Test Gap
0	Baseline (d0=64, d1=128, kf=64, θ_1=0.001)	1.4189	1.4593	0.33
2	Stronger reg (θ_1=0.1, θ_2=0.01)	1.3656	1.4098	0.31
3	Smaller model (d0=32, d1=64, kf=32)	1.4034	1.4449	0.32
4	Small model + strong reg	1.3735	1.4162	0.30
Final	**Hyperopt best (d0=64, d1=64, kf=128, lr=0.005, actv1=relu, θ_1=0.1, θ_2=0.1)**	**1.3619**	**1.4096**	**0.40**
Paper	MG-GAT [FR and NIG interpretable]	—	1.249	—

5.2 Comparison with Paper Baselines

Model	Test RMSE (Paper)	Test RMSE (Ours)
MG-GAT [interpretable]	1.249	1.4096
DGAN	1.250	—
GRALS	1.328	—
SVD++	1.339	—

Our result (1.4096) does not match the paper baselines directly, as the datasets differ significantly (320K vs 77K users). The gap is primarily attributable to the much sparser 2024 dataset (see Section 6).

5.3 User Filtering Ablation

Configuration	Train Reviews	Val RMSE	Test RMSE
Full training data	820,496	1.3616	1.4086
Filtered (≥2 ratings/user)	698,462	1.3729	1.4193

Filtering removed 70.3% of users (only 29.7% had ≥2 training ratings), degrading performance. Full data was used for the final model.

6. Gap Analysis

The gap between our result (1.4096) and the paper (1.249) is approximately **0.161**. We decompose it into four factors:

Factor	Est. Contribution	Evidence
Dataset sparsity (primary)	~0.08–0.10	70.3% of users have only 1 training rating; NIG ablation reversal (Section 7) confirms sparsity effect
Missing LLM perceptual map	~0.02–0.03	Paper Table 2: removing any graph type → ~0.03 RMSE increase

Insufficient HP search	~0.02–0.03	Explored 20/648 combinations (~3%); paper does not disclose search extent
Random seed / initialisation	~0.01–0.02	Stochastic variation
Total estimated	**~0.13–0.18**	**Covers the observed gap of 0.161**

The NIG ablation reversal provides the strongest evidence for the sparsity hypothesis: in our sparse data, uniform attention *outperforms* learned attention (1.4028 vs 1.4096), whereas in the paper's denser data, learned attention significantly helps (1.249 vs 1.303). This confirms that the attention mechanism requires sufficient per-node observations to learn meaningful weights.

7. Ablation Study [Section 5.3; Table 2]

Configuration	Implementation	Our Test RMSE	Paper Test RMSE
Full MG-GAT	—	1.4096	1.249
NIG Removed	$\alpha = 1/\|N_i\|$ (uniform weights)	1.4028	1.303
FR Removed	$H^{(1)} = \text{sigmoid}(W^{(1)} \cdot S)$	1.4111	1.305
Uniform Graph Weighting	$\omega_g = 1/3$ (fixed)	1.4042	1.280
No Auxiliary Information	S_u, S_b replaced by learnable embeddings	1.4390	1.312
No Networks or Auxiliary Info	Pure matrix factorisation	1.4640	1.405

FR Removed: RMSE increases (1.4096 → 1.4111), consistent with the paper. The linear Layer 1 design incurs a small performance cost for interpretability. [Appendix E, Eq.13]

NIG Removed: Unlike the paper (+0.054 degradation), our result shows a slight *improvement* (−0.007). This reversal is explained by data sparsity: with 70.3% single-rating users, the attention mechanism lacks sufficient signal to learn discriminative weights; uniform weighting acts as a stronger regulariser. This highlights an important interaction between data density and model complexity.

Uniform Graph Weighting: Fixing $\omega_g = 1/3$ for all business graphs yields 1.4042 vs 1.4096 for the full model. Similar to NIG Removed, the uniform variant slightly outperforms learned weights in our sparse setting, whereas the paper shows a clear benefit of learned weights (1.249 vs 1.280). This further confirms that data sparsity limits the model's ability to learn discriminative graph-level weights.

No Auxiliary Information: Removing side features S_u and S_b (replaced by learnable embeddings) increases Test RMSE to 1.4390, a degradation of 0.029 from the full model. This confirms that explicit user/business features provide valuable signal. The paper shows a larger degradation (1.249 → 1.312 = +0.063), suggesting that side features are even more valuable in the denser dataset.

No Networks or Auxiliary Information: The pure matrix factorisation baseline achieves Test RMSE 1.4640, the worst among all configurations. This confirms that both graph structure and side features contribute meaningfully to prediction quality. Notably, this model showed severe overfitting (Train 0.68 vs Test 1.46) and never triggered early stopping within 200 epochs, indicating that without structural regularisation from graphs, the model memorises training data.

8. Implementation Notes

Design Decision	Paper Basis
Graph regularisation only on $H^{(4)}$, not $H^{(3)}$	"H(3) is not regularized... because the graph attention design in Eq.(4) enables local smoothness" [Section 3.4]
3 business graphs stored independently, combined via ω_g	"w_g ∈ R represents a learnable parameter for each business

	graph g" [Section 3.3]
Loss computed only at observed positions	"Ω_training ∘ (X − X̂)" [Eq.8]
SVD uses training data only	Implied by time split [Appendix D2]
Layer 1: no bias, no activation	Eq.2 — linear for FR interpretability [Section 3.3]
Co-visitation graph from training reviews only	Implied by time split [Appendix D2]
θ_1 controls graph reg strength; θ_2 controls $\tilde{L}$ diagonal; weight_decay controls L2	Three separate roles [Section 3.4; Footnote 6]

9. Conclusion

We reproduced the core MG-GAT architecture (Stage 1) following the paper's specifications. The final model achieves a Test RMSE of **1.4096** compared to the paper's **1.249**. The implementation correctly captures: the five-layer interpretable architecture with FR and NIG; three business graph types with learnable ω_g weights; graph Laplacian regularisation on $H^{(4)}$ only; Hyperopt-based hyperparameter search; and full ablation study (6 configurations matching paper Table 2).

The ablation study confirms the value of each model component: graph structure and side features both contribute (removing either degrades performance); pure MF without graphs or features is the worst (Test RMSE 1.4640). The NIG and Uniform Graph Weighting ablations show reversed effects vs the paper due to data sparsity.

The performance gap (~0.161) is primarily attributable to dataset version differences — the 2024 Yelp dataset has 4.2× more users but much higher sparsity (70.3% single-rating users), which fundamentally limits the attention mechanism's ability to learn meaningful neighbor weights. This is directly evidenced by the NIG ablation reversal. Secondary factors include the excluded LLM perceptual map graph and limited hyperparameter search coverage (~3% of the space).